

Integration of Hybrid Energy Model with Solar PV, Hydro & Wind Turbine by Using MATLAB/Simulink

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Abstract— There are various sustainable energy sources which play vital role in producing feasible electricity with less gas emissions and have received extensive attention of researchers across worldwide. Sunlight and wind are two commonly used renewable energy sources. There are two important factors such solar irradiance and wind speed which may change in an unconditional manner. This further produces challenges with the integration of power grid because sometimes electricity may not produce as per load requirement. Hence instead of using single energy source hybrid energy system has been introduced for the production of continuity of electricity. Hybrid energy system may consist of renewable and non-renewable energy sources. In this paper integration of hybrid power system with solar PV cell, wind turbine and hydro energy system has been described under environment of MATLAB/Simulink and done comparative analysis of hybrid energy model.

Keywords— *solar pv cell, wind turbine, matlab, hydro energy, hybrid power system, mppt*

I. INTRODUCTION (HEADING 1)

In today's era electric power system contains of various equipments for different operation such as production, transmission and distribution. Because demand of electricity is increasing due to presence of natural energy sources as fossil fuel produce more gas emission and provide less reliable electricity. Sunlight and wind energy are two most commonly used energy sources. Solar irradiance and wind speed are two important factors on which electricity production depends. These provide smooth operation and get more attention and interest [1].

According to Energy report energy production has been increased at rate up to 1.6 percent per year and by year 2030 demand of energy will be 17000 Mtoe with an overall increment of 45 percent. In this momentous duration for future energy production and gas emission, natural gas demand is around 650 billion cubic meters lower in 2050 with less projected consumption in power.

Many authors have been proposed various hybrid energy models with new integration scheme through mostly for wind/photovoltaic cell comprises of both grid side and rotor side converters for injecting the power into grid [2]. This helps in improving efficiency of hybrid energy system.

Electricity is taking as center role in the lives of customers and increasing number of household because this becomes as base on which each operation is dependent either in household or in industry. Without electricity no

production is possible and cannot complete our daily needs. Hence through using single energy system sometimes it may fail due to presence of less environmental impact so hybrid energy system has been introduced [3].

Standalone PV systems are used to perform as independent source because this is behaving as directly coupled device. It converts sunlight into electricity which can be further used for various household applications. So this conversion is possible by using only some devices called solar photovoltaic panels. The produced power depends upon type of load which is connected in the system. In addition with tacking of voltage and current different MPPT techniques can be used. Some of authors have been proposed alternative transient program with electromagnetic transient and ATP draw versions for modeling and simulation [4]. There are different emerging technologies for designing of hybrid energy system in which energy source performs in a sequential manner for longer running of time [5]. The rate of using renewable energy has been increased from year 2012-2018 as demand increased [6]. During designing of hybrid power system strategy of energy management is also required if it is interconnected with electric vehicle through the use of fuzzy logic controller which in turns gives better energy storage capacity [7]. There are various factors such as power quality and voltage stability upon which efficiency can be improved with damping out oscillation in hybrid power system [8].

II. INCREMENTAL CONDUCTANCE METHOD

The problem of perturb and observe MPPT technique has been resolved by using incremental conductance method for better performance. It can determine the operating point on its maximum position and then stop next perturbing operation. If this condition is not verified then maximum power point must be perturbed can be evaluated by using relationship between dI/dV and $-I/V$. All these relations have been derived when dP/dV is negative when MPPT is to the right of maximum power point and positive when this is to the left of MPP. Further I&C method can track maximum power point rapidly and it can perform faster than P&O.

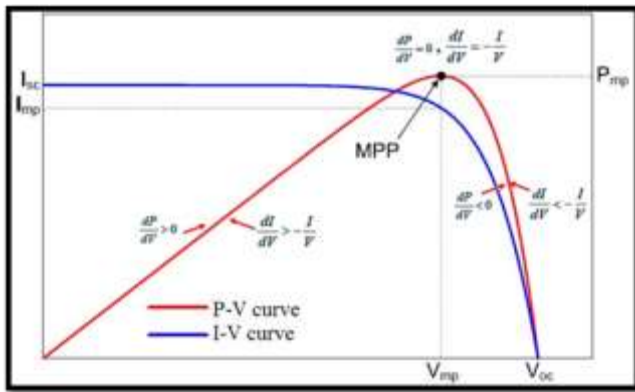


Fig. 1. Graph of I&C Method [9]

The basic equations can be written as mathematically from the graph:

$dI/dV = -I/V$ at maximum power point
 $dI/dV > -I/V$ at maximum power point
 $dI/dV < -I/V$ at maximum power point

III. HYBRID MODEL INTEGRATION WITH SOLAR PV CELL-WIND TURBINE

Hybrid Power system contains of two renewable source of energy such as solar PV cell & wind turbine with MPPT tracking technique used for extraction of maximum energy during the operation. Using this hybrid energy model voltage, current and power have been calculated using MATLAB.

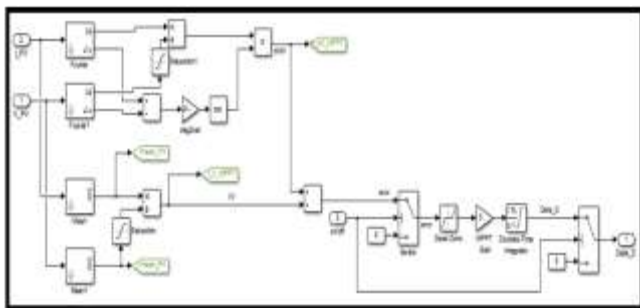


Fig. 2. MPPT Algorithm Subsystem Model

By using hybrid power source it is possible to generate energy on continuity basis. For solar PV cell solar irradiance is an important parameter but in winter season it is difficult to produce electricity by using solar PV cell. Same for wind turbine wind speed helps in production of electricity hence all these operation work at point of common coupling. Incremental conductance method is used for tracking of maximum power for this hybrid power system.

Wind turbine model comprises of generator which rotates to produce electricity. Generator is interconnected to the bridge rectifier which results in rectified dc output further applies to input of boost converter.

The mathematical expression of mechanical power which governs the rotation of electric generator:

$$P = \frac{1}{2} \rho A C_p V^3 \dots \dots \dots (1)$$

Where ρ is air density (kg/m^3), A is area swept by rotor blades, V is air velocity and C_p is power coefficient. Then tip speed ratio is defined by:

$$\text{TSR} = \omega_m R/V \dots \dots \dots (2)$$

Where R is radius of turbine and ω_m is mechanical angular speed with V as wind speed.

Below fig.3 is showing hybrid model of solar-wind energy system under MATLAB environment.

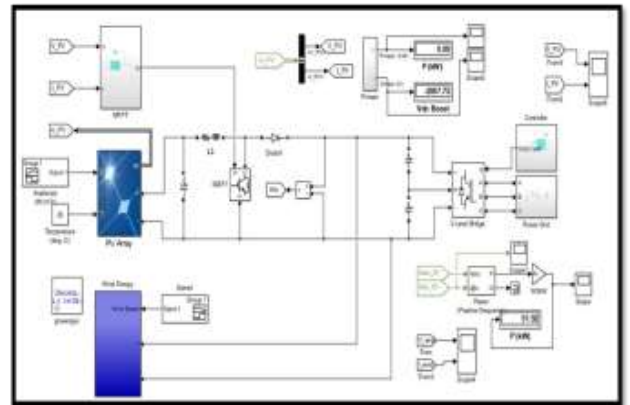


Fig. 3. Solar Wind Turbine Hybrid Energy Model

In this hybrid model two non-conventional sources have been modeled with I&C MPPT technique for extracting maximum amount of power. Here duty cycle has been taken for the operation of hybrid system. DC-DC converter has been employed for boosting maximum output power. Two important parameters such as temperature and solar irradiance are playing vital role for solar PV cell. In wind energy system wind speed is an important factor for the operation.

A DC-DC boost converter has been employed to convert unregulated DC input to controlled DC output. For continuous conduction mode higher switching frequency is required.

Some of hybrid energy models have been integrated based on wind and hydro energy system and during operation various numerical tests has been performed by using IEEE bus system[10].

IV. HYBRID MODEL INTEGRATION WITH SOLAR- HYDRO POWER SYSTEM

Hydro power is one of most widely used renewable energy sources which convert the energy from flowing water into electricity by using alternator. Turbine governor is a device which helps in regulating water input to turbine for the production of electricity.

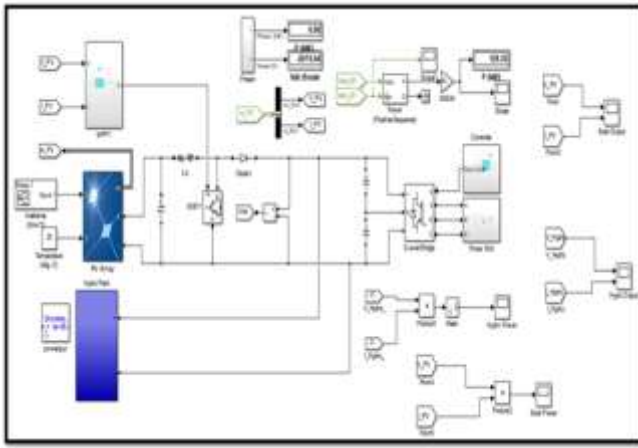


Fig. 4. Solar Hydro Power Hybrid Energy Model

In above hybrid energy model same I&C MPPT technique has been preferred for extracting maximum amount of power. Hydro energy model has based on Bernoulli equations on which it is working. There are three main components of hydro energy system such as governor, exciter and generator for production of electricity. There are different optimization techniques which can be used for load frequency and model predictive control and their results have been compared with other optimization techniques [11]. Some authors have been focused on challenges due to integration of renewable energy sources for hybrid energy system modeling through the initiation of hybrid micro-grid [12]. There are various tools available for evaluating the performance using power optimization software making cost effective hybrid energy model [13]. HOMER software is widely used tool for optimization of hybrid energy system considering output in the form of fuel cost, wind speed and solar isolation [14]. Feasibility is another important factor which can be analyzed on HOMER software tool with its fast achieving optimal solution and simplicity characteristics [15].

TABLE I. COMPARATIVE ANALYSIS OF OPTIMIZATION TECHNIQUES [16]

S.No	Name of Techniques	Proposed Limitations	Advantages
1.	Linear Programming	More cost	Adequate optimal issues
2.	Numerical Method	Requires graphical representation	Amended accuracy
3.	Probabilistic Methods	Reduced dynamic performance	Better operating characteristics
4.	Particle Swarm Optimization	Increased uncertainty	Easy calculation
5.	Ant Colony Optimization	Increased complexity	Less component size
6.	Artificial Bee Colony	More cost	Good convergence speed

V. SIMULATION RESULTS OF SOLAR-WIND TURBINE HYBRID MODEL

By employing hybrid energy system through the use of solar PV cell and wind turbine under MATLAB

environment here at one time one energy source is activating. In this model duty cycle controller has been used when solar PV cell operates then at the same time wind turbine will be deactivated. After reaching 0.5 sec wind turbine is operating. From fig. 5, 6 & 7 it is showing voltage, current and power of solar PV cell module. The voltage is showing around 250 v and current is 380 A and then it has been stopped.

Now after 0.5 sec wind turbine operates and below figure it is showing voltage and current curve. The voltage and current rating are around 500V and 200A.

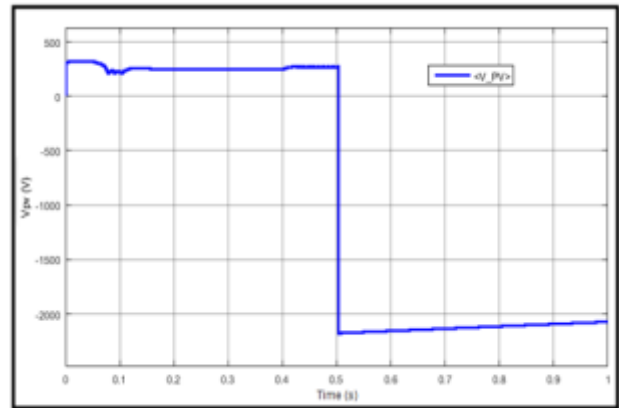


Fig. 5. Vpv Result

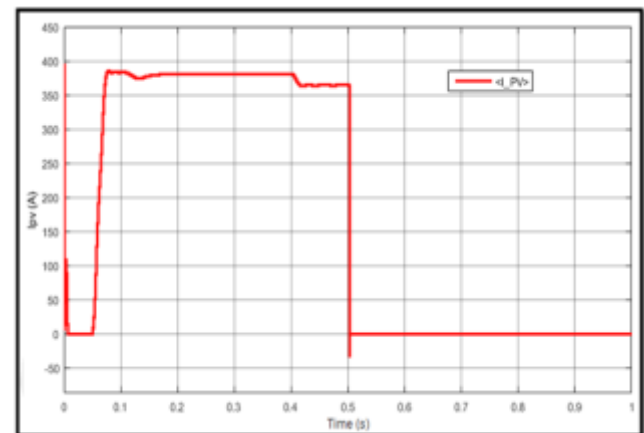


Fig. 6. Ipv Result

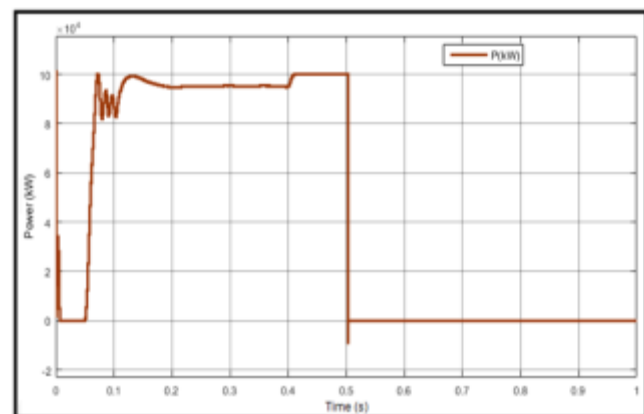


Fig. 7. Ppv Result

Fig. 8.

When time reaches 0.5 sec then wind turbine system will be activated and solar PV cell will be stopped working.

Then according to wind energy parameters we are getting results of voltage, current and power.

Below figure 8, 9 and 10 are showing results of voltage, current and power of wind turbine energy model in hybrid mode.

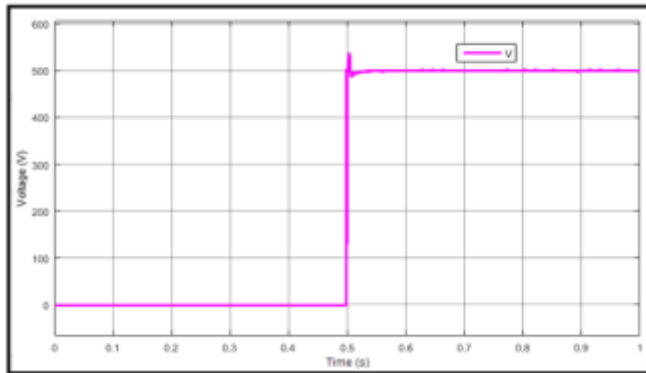


Fig. 9. Wind Voltage Result

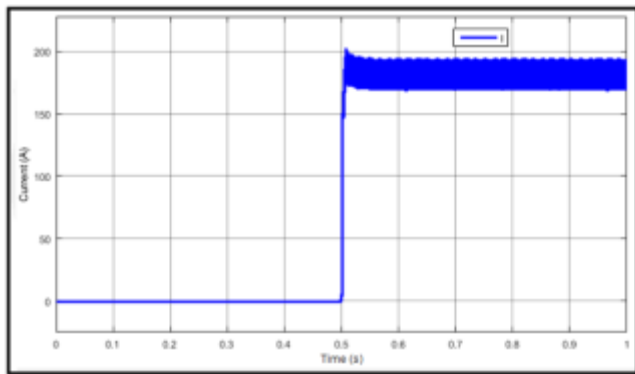


Fig. 10. Wind Current Result

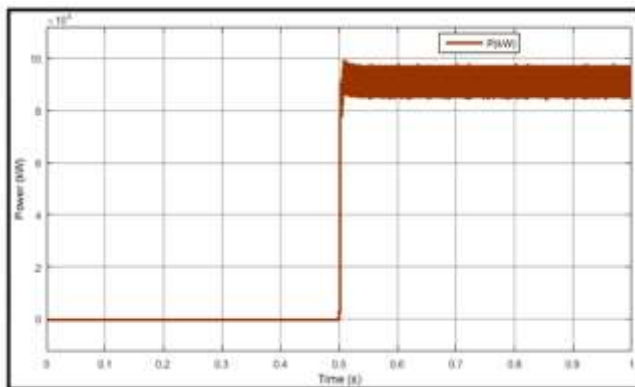


Fig. 11. Wind Power Result

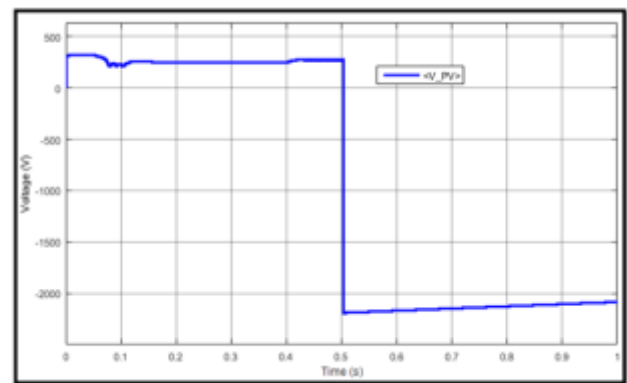


Fig. 12. Vpv Result

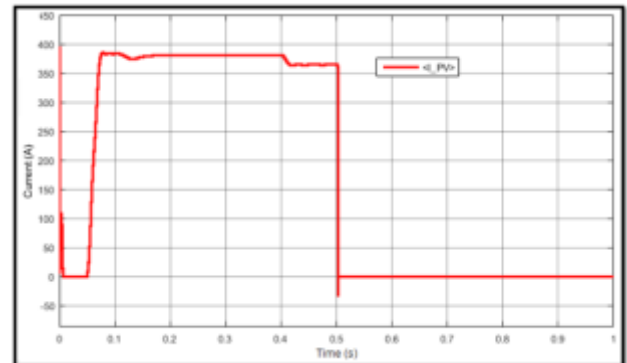


Fig. 13. Ipv Result

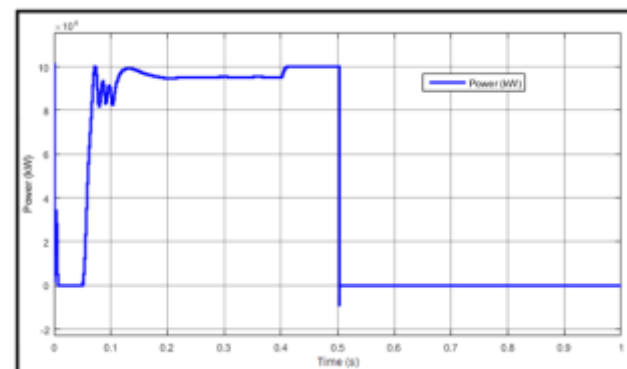


Fig. 14. Ppv Result

After 0.5 sec hydro energy model is working and figure 14, 15 and 16 are showing results of hydro energy model in hybrid mode. In hydro energy model through the use of various components it is showing some harmonics but it reaches to stable state.

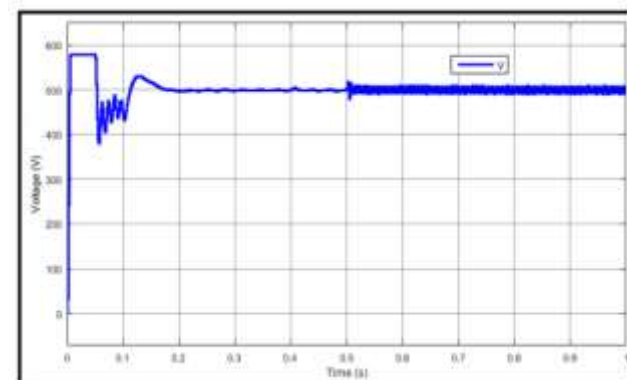


Fig. 15. Voltage of Hydro Power Plant

VI. SIMULATION RESULTS OF SOLAR-HYDRO POWER HYBRID MODEL

Same configuration has been integrated of solar and hybrid energy system. Firstly solar PV cell is working and fig. 11, 12 and 13 are showing results of solar PV cell.

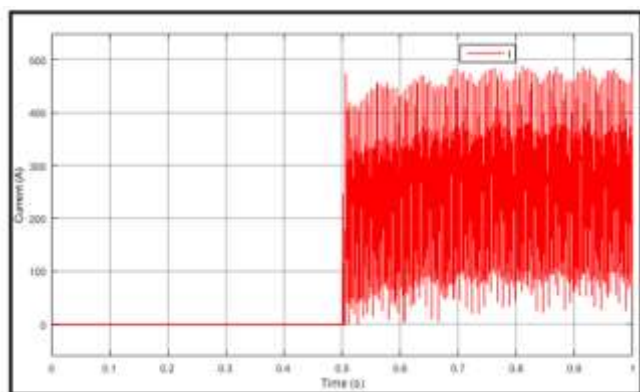


Fig. 16. Current of Hydro Power Plant

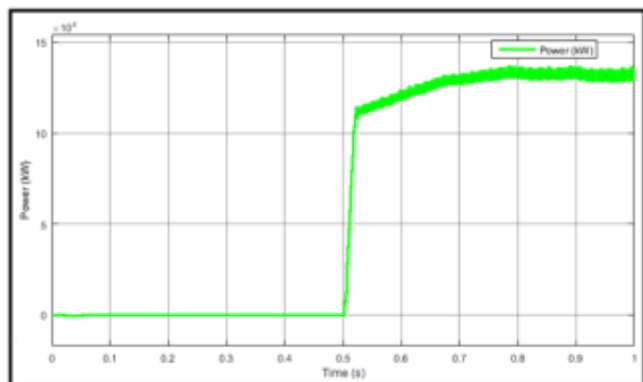


Fig. 17. Power of Hydro Power Plant

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CONCLUSION

In this paper integration of hybrid energy system by using solar PV, wind and hydro power has been analyzed through the use of MPPT technique. Then we have compared two different hybrid energy models under MATLAB environment and determined the best operation of proposed hybrid model. Through the simulation results solar-wind hybrid model gives better results rather than solar-hydro hybrid energy hybrid model which may further used for determining its stability and reliability. The novelty of this paper is to analyze characteristics of parameters of hybrid energy system for achieving stable operation as in order to reduce harmonics in a system. This work can be further used for determining feasibility operation with electric vehicle under different charging conditions.

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